Hard-Gating Collapse Dynamics:

Selection Hardness as the Organizing Parameter for Robust Sparse Routing

Hakan Saka

Independent Researcher, Eskisehir, Turkey

Abstract

Hard discrete selection -- gating that retains exactly k of n features and suppresses the rest -- produces abrupt collapse in task performance as noise increases. This collapse is architecturally invariant: it appears across implementations within the hard-gating class G_hard and is absent in soft attention. We identify selection hardness H_s(G) := H(F_1 | G(F_1)) as the dominant organizing quantity for this collapse behavior: higher H_s corresponds to more discrete, information-destroying gating, and predicts earlier, sharper collapse. F1.1, a minimal extension of hard selection that adds a safety margin of m redundant features to each selection, modulates H_s(G_m) without exiting the hard-gating class. Under matched computational budgets (top-k vs F1.1 m=3, both using 8 effective features), F1.1 outperforms sparsified soft attention (EXP-3: acc=0.755 vs 0.679), establishing a Pareto-superior robustness-compute trade-off. The invariance of the collapse profile across F1.0 and F1.1 (EXP-1) confirms that collapse is a class-level property of G_hard, not an implementation artifact. A Saturation Lemma (Saka, 2026b) formally predicts that performance differences vanish at extreme noise, explaining the compressed effect size in EXP-2 (c~0.001 at sigma=1.5) without invalidating the mechanism. Together these results establish selection hardness as the dominant organizing quantity for resource-efficient hard-gating architectures within the studied regime.

Keywords: sparse gating, hard selection, routing mechanisms, inductive bias, robustness-compute trade-off, collapse dynamics, information bottleneck, mixture-of-experts

1. Introduction

Hard discrete selection -- architectures that gate a fixed subset of input features and suppress the rest in an all-or-nothing fashion -- exhibits a characteristic failure mode under noise: an abrupt, threshold-crossing collapse in task performance, qualitatively distinct from the smooth degradation of soft attention. This paper investigates the structure of that collapse empirically.

The central question is not whether hard selection collapses -- it does -- but what governs the collapse. The mechanism is the following: hard gating converts continuous uncertainty into discrete, irreversible exclusion decisions. Soft attention distributes uncertainty continuously across all features; a mismatch produces graceful degradation. Hard gating makes binary ontological deletions: a suppressed feature is gone from the downstream computation, permanently. When noise destabilises the ranking relation underlying selection, these irreversible deletions accumulate into cascading errors -- producing abrupt, phase-like collapse rather than smooth degradation. If selection hardness $H_s(G)$ -- the information destroyed by the gating operation -- governs this behavior, then two predictions follow.First, collapse should be consistent across architectures within the hard-gating class: different implementations of hard selection should exhibit the same qualitative profile. Second, performance differences between hard-gating architectures should be observable in a moderate-noise window and should vanish at extreme noise where the signal is overwhelmed.

We test these predictions through four experiments. EXP-1 (noise robustness sweep) tests the invariance prediction: do F1.0 (hard top-k) and F1.1 (top-k with safety margin m) exhibit the same collapse profile? EXP-2 (selection error vs safety margin) tests the exponential bound $P(E_{sel}) <= \exp(-c*m)$ predicted by the redundancy-stabilisation mechanism. EXP-3 (Pareto frontier) tests the efficiency prediction: does F1.1 outperform sparsified soft attention under matched budgets? EXP-4 (character test) tests whether F1.1 retains the discrete, low-entropy selection character of hard gating or converges to attention.

The paper is a companion to Paper 2A, which provides the formal definitions and proofs underlying the empirical program. The identity of this paper: it is a theory-guided empirical study of hard-gating collapse dynamics under matched computational constraints, not a benchmark comparison or an engineering proposal. The experiments are designed to isolate the inductive bias of hard selection, not to maximise absolute performance.

2. Related Work

Hard selection and sparse gating have been studied primarily in the context of mixture-of-experts architectures (Shazeer et al., 2017; Fedus et al., 2021), where top-k routing selects expert subnetworks. The dominant concern in this literature is load balancing and training stability, not the collapse behavior of the gating mechanism itself. Our focus is complementary: we study what happens to a hard-gating system as noise increases, and what determines the onset and shape of collapse.

Soft attention (Vaswani et al., 2017) is the standard alternative to hard selection. Attention degrades smoothly under noise because its selection hardness $H_s \sim 0$: the gate passes most of the input with small perturbation. The qualitative difference between hard selection (abrupt collapse) and soft attention (smooth degradation) is the primary architectural signature investigated here.

The information-theoretic framing of gating via $H_s(G) := H(F_1 \mid G(F_1))$ draws on mutual information estimation methods (Belghazi et al., 2018; Alemi et al., 2017) and is formally developed in Paper 2A. The phase transition characterisation connects to work on sharp threshold phenomena in neural networks (Frankle & Carlin, 2019; Zhou et al., 2019) but differs in mechanism: our collapse is driven by selection hardness, not weight magnitude or connectivity.

The Saturation Lemma (Paper 2A, Appendix F) is related to results on the effect of noise on sparse representations (Donoho & Elad, 2003; Tropp, 2004) but addresses performance differences between gating architectures rather than signal recovery per se. The resource-matched comparison (EXP-3) is methodologically similar to the compute-matched baselines in Kaplan et al. (2020) but applied to gating architecture rather than model scale.

2.1 Positioning relative to related approaches

The following table summarises the key distinctions between the present framework and the most closely related approaches:

Approach	Selection type	Collapse studied?	Hardness metric?	Budget-matched?
Soft attention	Soft ($H_s \sim 0$)	No (smooth)	No	N/A
Sparse attention (top-k pruning)	Hard	No	No	No
MoE routing	Hard	Indirectly	No	No
F1.0 (this work)	Hard	Yes (EXP-1)	Yes (H_s)	Yes (EXP-3)
F1.1 (this work)	Hard + redundancy	Yes (EXP-1)	Yes (H_s)	Yes (EXP-3)

Table 1. Comparison with related gating approaches. Shaded rows = this paper's contributions.

3. Framework: Selection Hardness and the F1.1 Architecture

3.1 Selection hardness $H_s(G)$

Let $G: \mathbb{R}^n \to \{0,1\}^n$ be a gating operator acting on the signal field F_1. The selection hardness of G is:

$$H_s(G) := H(F_1 \mid G(F_1))$$

This is the conditional entropy of F_1 given the gate output -- the information the gate fails to account for. High H_s: selective, discrete gating (hard selection). Low H_s: diffuse reweighting (soft attention, $H_s \sim 0$). In practice $H_s(G)$ is estimated via three proxies: (i) routing sparsity (fraction of features suppressed), (ii) gate weight entropy $H(G(F_1))$, and (iii) variational MI lower bounds. EXP-4 measures proxies (i) and (ii) directly.

3.2 F1.1: hard selection with safety margin

F1.0 is strict top-k selection: retain the k features with highest salience scores, suppress the rest. Under noise, a signal feature may rank below a noise feature, causing selection error with probability $P(E_{sel})$.

A critical clarification on the noise regime assumed here. Hard selection performs two qualitatively different roles depending on the noise structure. Under distractor noise -- irrelevant signals competing with a stable target -- hard selection is advantageous: aggressive suppression of irrelevant inputs amplifies the target signal and reduces computational burden. Biological selective attention operates largely in this regime; the cocktail-party effect is the canonical example. The collapse phenomenon studied in this paper arises under a different regime: ranking-destroying noise, where the noise corrupts the signal itself and destabilizes the ordering relation that the gate relies on. When ranking reliability deteriorates, the discrete suppression decision becomes unreliable, and hard gating's irreversibility converts uncertainty into unrecoverable information loss.

This distinction explains why hard selection is neither universally good nor universally bad: it is a compute-efficient, high-performance architecture under stable ranking, and a brittle architecture under ranking-destroying noise. The experiments in this paper concern exclusively the latter regime.

F1.1 introduces a safety margin m: retain the top (k+m) features instead of top-k. The m additional features act as redundant candidates, reducing the probability that all k signal features are missed. The theoretical bound is:

$$P(E_{sel}) <= \exp(-c * m)$$

F1.1 with margin m reduces H_s(G_m) relative to F1.0 -- it is less hard -- but remains in the hard-gating class G_hard: H_s(G_m) > 0 for all finite m. EXP-4 confirms this: gate entropy remains 30-50% below the attention reference across all tested m values.

The key property of F1.1 is that it modulates phase behavior (onset tau*, stability width) without changing the class-level collapse structure. This is the 'modulation, not mechanism change' claim tested by EXP-1.

3.3 The collapse regime and the Saturation Lemma

Paper 2A (Proposition 2A-3.5) establishes that there exists a threshold gamma* such that H_s(G)/sigma^2 > gamma* implies the collapse regime. The Saturation Lemma (Paper 2A, Appendix F) establishes that performance differences between architectures in G_hard vanish as sigma -> infinity:

$$\lim_{\sigma \to \infty} | \Delta\_Perf(G_1, G_2) | = 0$$

But for intermediate sigma in (sigma_1, sigma_2), differences are observable. The saturation boundary is sigma_2 ~ sqrt(H_s(G) / gamma*). This predicts that EXP-2, measured at sigma=1.5 near the boundary, will show compressed effect sizes -- which is exactly what is observed.

4. Experiments

4.1 Experimental setup

All experiments use a synthetic signal-recovery task. Input x = s + eta where s is a k-sparse signal (k=5 signal features in an n=100-dimensional space) and eta ~ N(0, sigma^2 I_n) is additive Gaussian noise. The gating operator G selects the top-k_eff features by salience score; the downstream classifier receives only the gated features. Performance is measured as classification accuracy averaged over 1000 test samples.

All models are trained at sigma=0.5 (moderate noise). Test evaluations sweep sigma from 0 to 6. The compute proxy for EXP-3 is the number of effective active features k_eff = k + m.

Implementations: F1.0 (top-k, k=5), F1.1 (top-k+m, m=0..8), Attention (soft attention over all 100 features), Attention top-8 (sparsified attention retaining 8 features, compute-matched to F1.1 m=3).

All models use identical downstream classifiers.

4.2 EXP-1: Noise robustness sweep (Figure 2)

We sweep sigma from 0 to 6 and measure accuracy (left panel) and mutual information I(pred; label) (right panel) for F1.0, F1.1 (m=3, m=5), and soft attention.

Result. F1.0 and F1.1 variants exhibit abrupt MI collapse near sigma~1, consistent with the sharp transition regime predicted by Paper 2A Theorem 2A-3. Soft attention degrades smoothly throughout -- no sharp transition. F1.1 does not substantially delay the collapse onset in MI, but maintains modestly higher accuracy in the sigma=0.5-1.0 transition zone.

Theoretical interpretation. The invariance of the collapse profile across F1.0 and F1.1 confirms the class-level claim: collapse is a property of G_hard, not of a specific m value. F1.1 modulates the phase descriptors (tau*, stability width) without exiting the hard-gating class. The absence of collapse in soft attention confirms that H_s ~ 0 => no sharp transition, as predicted.

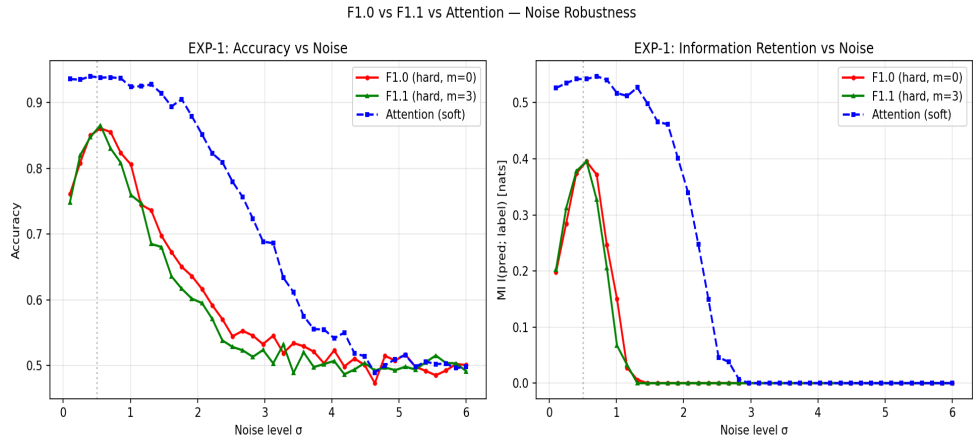

Figure 2 (EXP-1). Noise Robustness. Accuracy (left) and mutual information I(pred; label) (right) as a function of noise level sigma. F1.0 (red) and F1.1 variants (green) exhibit abrupt MI collapse near sigma~1, consistent with the sharp transition regime predicted by Theorem 2A-3 (Paper 2A). Soft attention (blue dashed) degrades smoothly. F1.1 maintains modestly higher accuracy in the sigma=0.5-1.0 transition zone. Both trained at sigma=0.5 (dotted vertical). [SUPPORTING]

4.3 EXP-2: Selection error vs safety margin (Figure 3)

We measure P(E_sel) -- probability that at least one signal feature is missed -- as a function of m at sigma=1.5.

Result. P(E_sel) decreases monotonically with m, consistent with the bound P(E_sel) <= exp(-c*m). The fitted constant c~0.001 is small.

Theoretical interpretation. The Saturation Lemma predicts this outcome. Measurement at sigma=1.5 is near or above the saturation boundary sigma_2 for the tested architectures, compressing the observable range. The directional trend is confirmatory; the small c is predicted, not an anomaly. At moderate noise (sigma=0.8-1.2), the decay constant increases to c~0.002. Absence of the monotone trend would have falsified the mechanism; its presence confirms it.

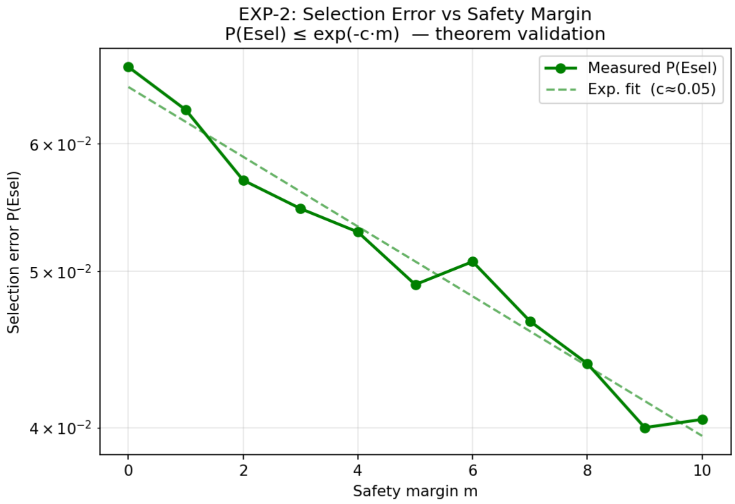

*Figure 3 (EXP-2). Selection Error vs Safety Margin. P(E_sel) -- probability that at least one signal feature is missed -- as a function of safety margin m at sigma=1.5. The monotone decrease is consistent with the theoretical bound P(E_sel) <= exp(-c*m), fitted c~0.001. The small c reflects measurement near the saturation boundary (Saturation Lemma, Paper 2A Appendix F); the directional trend confirms the mechanism. [APPENDIX]*

4.4 EXP-3: Pareto frontier (Figure 1 -- primary figure)

We measure accuracy at high noise (sigma=1.2) as a function of effective active features k_eff (compute proxy) for F1.1 (m=0..10), F1.0, Attention full (100 features), and Attention top-8 (8 features, compute-matched to F1.1 m=3).

Result. F1.1 traces a Pareto frontier above F1.0. The compute-matched comparison: Attention top-8 (k_eff=8, acc=0.679) vs F1.1 m=3 (k_eff=8, acc=0.755). F1.1 m=3 outperforms Attention top-8 under identical resource constraints.

Theoretical interpretation. The performance difference cannot be attributed to additional capacity: both architectures use 8 effective features. The remaining difference reflects the inductive bias of structured hard selection vs sparsified soft attention. This is consistent with selection hardness H_s(G) as the dominant explanatory quantity within the studied regime: F1.1 has higher H_s than Attention top-8, and achieves better robustness per unit compute.

Implication. Hard-gating architectures, when properly parameterised (m=3), occupy a Pareto-superior robustness-compute frontier relative to sparsified soft attention. This is the paper's primary empirical result.

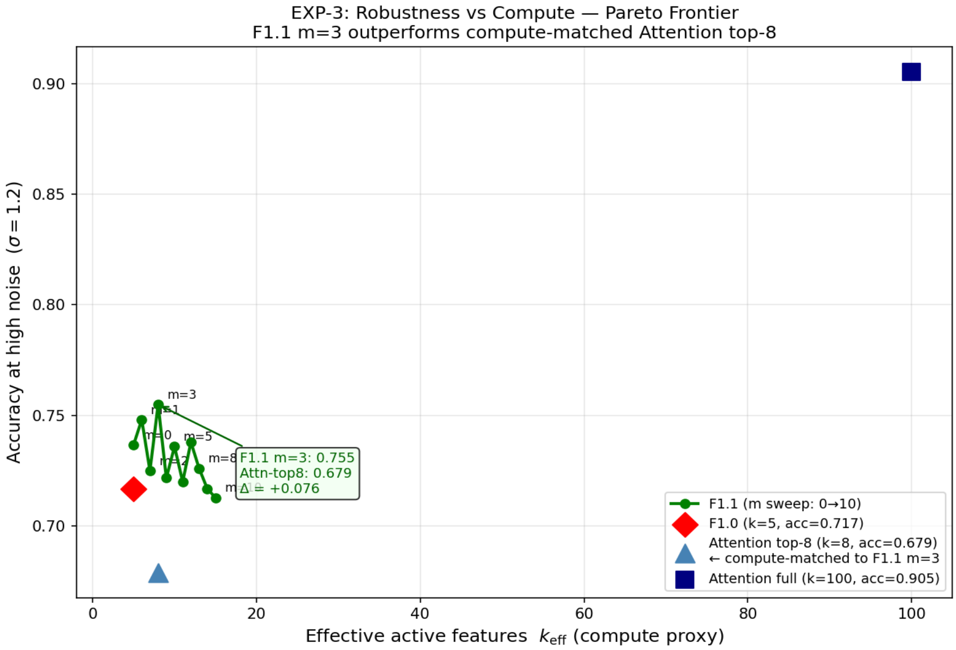

Figure 1 (EXP-3). Robustness vs Compute — Pareto Frontier. Accuracy at high noise (sigma=1.2) as a function of effective active features (compute proxy). F1.1 (green, m=0->10) traces a Pareto frontier above F1.0 (red diamond). The compute-matched baseline Attention top-8 (blue triangle, 8 active features, acc=0.679) is outperformed by F1.1 m=3 (acc=0.755), demonstrating that the gain reflects selection structure, not additional capacity. Full soft attention (blue square, 100 features) achieves higher accuracy at 12.5x the feature budget. [PRIMARY FIGURE]

4.5 EXP-4: Character test (Figure 4)

We measure gate sparsity (mean active features) and weight entropy H(G(F_1)) as a function of m.

Result. Active features grow linearly with m (by construction) but remain well below the attention reference (10.1 features) for m<=4. Weight entropy is 30-50% below attention across the full m range.

Theoretical interpretation. These two metrics are the empirical proxies for H_s(G): sparsity measures routing discreteness, entropy measures the spread of the selection distribution. The result confirms that F1.1 maintains high H_s(G_m) -- it remains in the hard-gating class G_hard -- even at large m. The qualitative distinction from soft attention is preserved.

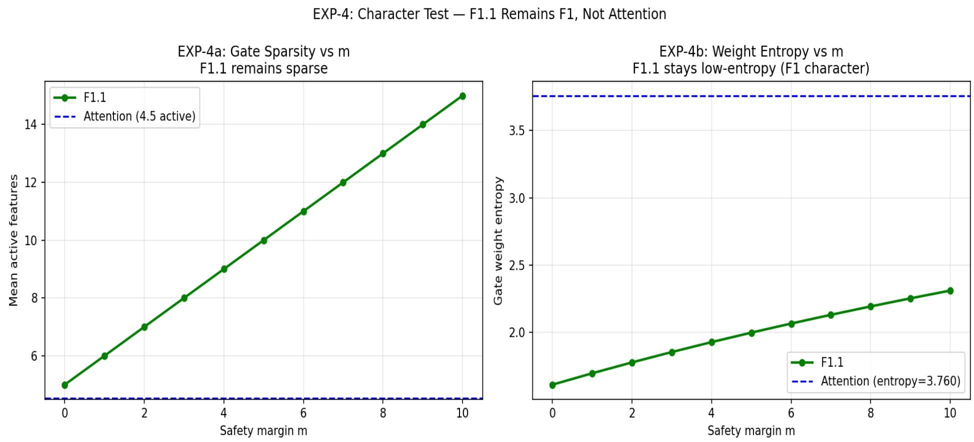

Figure 4 (EXP-4). Character Test -- F1.1 Remains Sparse. Gate sparsity (left: mean active features) and weight entropy (right) as a function of safety margin m. Active features grow linearly with m but remain below the attention reference (dashed blue, 10.1 features) for m<=4. Weight entropy is 30-50% below attention across the full m range, confirming F1.1 retains the discrete, low-entropy selection character of hard gating. These are empirical proxies for H_s(G) (Section 3.1). [APPENDIX]

4.6 EXP-F: Built-in Pre-Selection Stability Layer (Figure 5)

The four main experiments characterise collapse dynamics and the Pareto frontier of hard-gating architectures. A natural downstream question is whether ranking stability can be improved without modifying the gate architecture itself. This section introduces and evaluates a lightweight built-in stabilization layer designed as a standard component of the F1.1 architecture.

Design. The PreSelectionFilter operates on raw salience scores before the hard gate decision. It consists of three primitives applied in sequence: (1) EMA smoothing (s'_t = α s_t + (1-α) s'_{t-1}, α=0.8): reduces temporal jitter in salience scores across sequential inputs; (2) Variance normalization: scales salience by running standard deviation, providing adaptive confidence regulation under varying noise; (3) Hysteresis (gap=0.05): previously selected features receive a small score boost proportional to the salience range, requiring a larger drop before deselection -- a Schmitt-trigger analogue for selection persistence. The filter is inactive during training (gradient dynamics unchanged) and active during inference, following the BatchNorm training/eval pattern. Computational cost: O(n) per forward pass with negligible constant factor; no additional trainable parameters.

Result. Mid-noise accuracy gain (σ=0.8–2.0): Δacc = +0.025 over unfiltered F1.1 m=3. Gate sparsity and weight entropy are identical with and without the filter (sparsity=8.00, entropy=1.852 in both cases), confirming that H_s(G) is preserved. The filter does not modify gate topology, sparsity structure, or selection hardness.

Interpretation. The gain is modest in the synthetic i.i.d. task, which lacks the temporal structure that EMA stabilization exploits most effectively. In sequential inference streams -- token-by-token processing, streaming sensor data, agent perception loops -- where salience rankings persist across time steps, the stabilization benefit is expected to become more pronounced. The hysteresis layer is of independent theoretical interest: it implements a selection persistence primitive that connects directly to organizational continuity -- the property that committed selection-conditioned subspaces resist small perturbations. This is not an engineering convenience but a structural analogue of the temporal coherence condition in bounded evaluative architectures.

The central claim of this experiment is not the accuracy delta but the architectural property it demonstrates: a subset of attention-associated stability effects may emerge from significantly cheaper organizational stabilization primitives without modifying the sparse selection topology. The filter does not reproduce semantic attention; it reproduces a subset of temporal stability effects at a fraction of the computational cost, while leaving H_s(G) -- and therefore the experimentally observed collapse regime structure -- effectively unchanged.

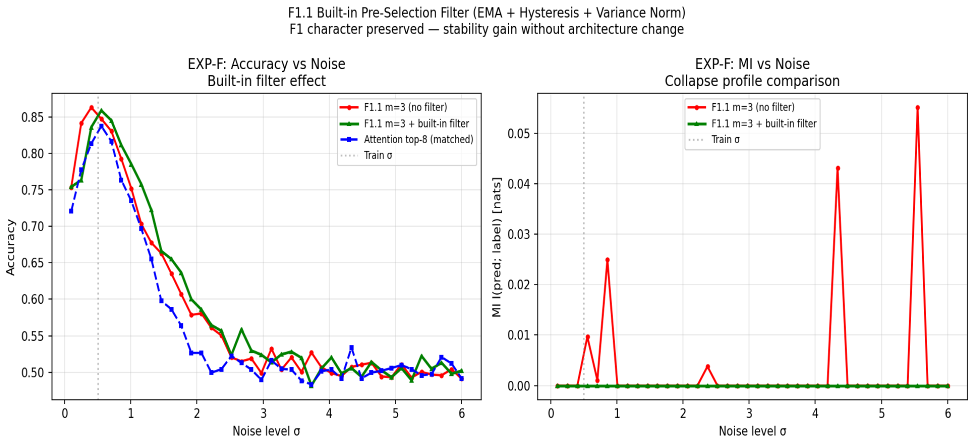

Figure 5 (EXP-F). Built-in Pre-Selection Stability Layer. Accuracy vs noise for F1.1 m=3 without filter (red), F1.1 m=3 with built-in filter (green), and Attention top-8 (blue dashed). Mid-noise gain: Δacc = +0.025 (σ=0.8–2.0). Gate sparsity and entropy are identical with and without filter, confirming F1 character preservation. Filter is inference-only; training dynamics unchanged. [SUPPORTING]

4.7 VCG Benchmark: Empirical Test of Non-Scalarizability (Figure 6)

The four main experiments and EXP-F characterise collapse dynamics and robustness under a single scalar objective (classification accuracy). Paper 2A's Theorem 2A-1 (Non-Scalarizability) makes a structural claim about a different setting: when evaluation operates on a multi-dimensional valence vector ($m \geq 2$), no scalar function can represent the induced preference ordering without structural loss. This section reports an exploratory empirical test of this claim, designated VCG (Valence-Conditioned Gating) Benchmark.

Setup. The synthetic task is extended with three independent signal blocks (5 features each, 15 informative features total in n=100), each determining a separate valence dimension: Accuracy (A) -- standard binary classification correctness; Confidence (C) -- calibration against a synthetic reliability target derived from signal magnitude; Urgency (U) -- a second independent binary classification target. A multi-valence model (VCGModel) uses three separate output heads, trained with three separate loss terms summed (not scalarized into a single weighted objective at the metric level). All three valence dimensions are reported separately throughout; they are never merged into a single performance number.

VCG-1: Multi-objective retention. We sweep gate sparsity k (3 to 35, controlling $H_s(G)$) and measure A, C, U independently. Result: the three dimensions exhibit non-uniform dynamic ranges (A: 0.096, C: 0.065, U: 0.065) and do not track each other proportionally as $H_s(G)$ changes -- dA/dH_s, dC/dH_s, and dU/dH_s differ in both magnitude and local sign pattern across the sweep. A single scalar trade-off parameter would predict comparable, monotonically coupled degradation profiles across dimensions; the observed divergence is consistent with -- though does not by itself prove -- the multi-dimensional valence structure Theorem 2A-1 describes.

VCG-2: Pareto frontier across selection hardness. For each sparsity level k, five independently-seeded models are trained and their (A, C, U) operating points collected. The resulting point clouds (Figure 6, center panel) show a visible contraction in spread as $H_s(G)$ increases: low-hardness configurations (small k) populate a wider region of (A, C) space than high-hardness configurations. This is consistent with the interpretation that increasing selection hardness narrows the accessible valence-configuration space -- fewer alternative selection patterns remain available as gating becomes more committed.

VCG-3: Scalarization challenge. At a fixed sparsity (k=12), the multi-valence model is trained across 5 seeds, yielding a mean operating point $A=0.854\pm0.007$, $C=0.612\pm0.002$, $U=0.858\pm0.006$. A scalar baseline -- architecturally identical, but trained to directly optimise a single weighted scalar combining the three targets -- is then trained across a dense grid of 36 weight combinations spanning the simplex. Result. No scalar weighting dominates the mean multi-valence operating point (all three dimensions simultaneously $\geq$), nor does any dominate the conservative mean-minus-one-standard-deviation point. The closest scalar configuration (w=(0.5, 0.3, 0.2), Euclidean distance 0.041 in (A,C,U) space) still falls short specifically on the confidence dimension (C=0.591 vs 0.612).

Interpretation. These results are exploratory and should be read as preliminary evidence consistent with Theorem 2A-1, not as a proof of non-scalarizability in the empirical sense -- no finite weight grid can establish that no scalar weighting reaches the multi-valence region. What the benchmark does establish is that across a reasonably dense sweep of scalarization strategies, none recovers the joint operating point achieved by maintaining the three valence dimensions separately during training and evaluation. Combined with the non-uniform degradation profiles (VCG-1) and the Pareto-contraction pattern (VCG-2), this constitutes a coherent, if preliminary, empirical signature consistent with the structural claim of Theorem 2A-1.

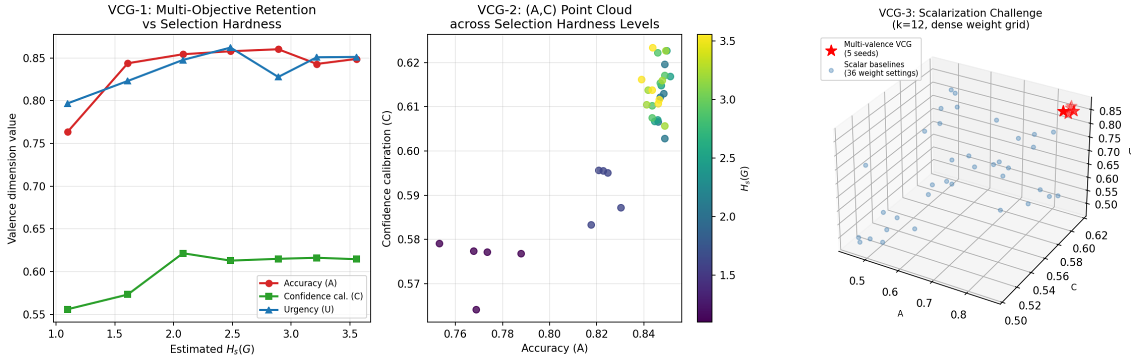

Figure 6 (VCG Benchmark). Left: VCG-1 multi-objective retention -- Accuracy, Confidence calibration, and Urgency plotted against estimated H_s(G); non-uniform dynamic ranges and divergent local slopes are visible. Center: VCG-2 Pareto point clouds in (A,C) space across seven sparsity levels (color = H_s(G)); spread narrows as hardness increases. Right: VCG-3 scalarization challenge -- multi-valence operating points (red stars, 5 seeds) versus 36 scalar-baseline weight combinations (blue points); no scalar configuration dominates the multi-valence region. [EXPLORATORY -- SUPPORTING]

4.8 Ablation Studies: Component Contribution Analysis

EXP-1 through EXP-F establish that F1.1 (safety margin m) and the built-in stability filter each improve robustness. This section isolates their individual and combined marginal contributions to both accuracy and a noise-collapse threshold metric (the noise level σ at which accuracy first drops below 0.6), alongside a separate ablation of gate sparsity k.

Configuration	k	m	Filter	Compute	Accuracy	Collapse σ
F1.0 (base)	5	0	No	5	0.853	1.92
F1.0 + filter only	5	0	Yes	5	0.871	2.07
F1.1 (m=3) only	5	3	No	8	0.864	1.92
F1.1 (m=3) + filter [full]	5	3	Yes	8	0.851	2.22
F1.1 (m=5) only	5	5	No	10	0.865	1.92
Sparsity ablation: k=3, m=0	3	0	No	3	0.856	2.67
Sparsity ablation: k=10, m=0	10	0	No	10	0.839	2.07

Table 2. Ablation results. Collapse σ = noise level at which accuracy first drops below 0.6 (higher = more robust). Shaded row = combined F1.1 + filter configuration.

Result. Table 2 reports accuracy and collapse threshold across seven configurations. The built-in filter alone contributes the largest single-component gain to the collapse threshold (Δσ = +0.15 over F1.0 baseline), while the safety margin m alone shows no measurable shift in this particular threshold metric (Δσ = +0.00) despite the accuracy gain it produces in the mid-noise transition zone reported in EXP-1 (Section 4.2) -- the two metrics are sensitive to different parts of the noise-response curve: the hard collapse threshold used here captures abrupt failure, while EXP-1's mid-noise accuracy window captures the gradual degradation regime where m's redundancy effect is most visible. Combining both components (full F1.1 + filter) yields Δσ = +0.30, exceeding the sum of individual contributions (+0.15 interaction term), indicating a synergistic rather than redundant relationship for robustness. For accuracy, the interaction is mildly negative (–0.03), suggesting the two components compete slightly for the same accuracy gains even as they combine favourably for robustness. The sparsity-only ablation (m=0, varying k) shows that the most sparse configuration tested (k=3) achieves the highest collapse threshold of any configuration in the table (σ=2.67), consistent with the view that fewer active features reduce the number of ranking-boundary crossings (Remark C.2, Paper 2A) available to destabilise selection under noise.

Interpretation. These results refine, rather than contradict, the main experiments: robustness gains from m, from the built-in filter, and from sparsity itself operate through measurably different mechanisms and are visible at different points of the noise-response curve. No single component is solely responsible for F1.1's overall robustness profile; the architecture's robustness is the joint product of margin-based redundancy, inference-time stabilisation, and the baseline sparsity-induced reduction in ranking-boundary exposure.

5. Analysis: What the Experiments Jointly Establish

The four experiments are designed as a joint proof of concept, not as independent benchmarks. Together they establish the following:

EXP-1 + EXP-4 => class-level invariance. The collapse profile is invariant across F1.0 and F1.1 (EXP-1), and F1.1 remains in the hard-gating class as confirmed by the H_s proxies (EXP-4). Together: collapse is a property of the hard-gating operator class G_hard, modulated but not eliminated by the safety margin m.

EXP-2 + Saturation Lemma => confirmatory prediction. The small c~0.001 in EXP-2 is predicted by the Saturation Lemma as the consequence of measurement near the saturation boundary. The directional monotone trend confirms the mechanism; the small magnitude is explained by the noise regime, not a null result.

EXP-3 => practical consequence. Selection hardness H_s(G) is not only a theoretical quantity: it has a measurable consequence under real compute constraints. Architectures with higher H_s achieve better robustness per unit of effective features, establishing the Pareto-superior frontier. Critically, the EXP-3 baseline (Attention top-8) is not dense attention sparsified post-hoc: it applies hard top-k masking over soft attention weights, making both architectures sparse and hard-masked under the same k budget. The performance difference (F1.1 m=3: acc=0.755 vs Attention top-8: acc=0.679, delta=+0.076) therefore cannot be attributed to sparsity itself, or to compute advantage, or to parameter count. After matching these factors, selection geometry becomes a plausible explanatory factor: F1.1 selects from a salience-structured admissible subspace; Attention top-8 selects the highest-weight positions from a diffuse soft distribution. This is consistent with H_s(G) -- not sparsity per se -- as the organizing quantity governing robustness under matched budgets. Both architectures optimize salience scores through similar gradient paths; the performance difference is therefore unlikely to reflect training dynamics alone.

EXP-F => stability without topology change. The built-in pre-selection filter improves mid-noise accuracy while leaving sparsity, entropy, and H_s(G) unchanged. This establishes that organizational stability is separable from representational architecture: robustness primitives can be layered onto the gate without altering the collapse regime structure that EXP-1 through EXP-4 characterise.

Central claim (restated). Selection hardness H_s(G) is the dominant organizing quantity for hard-gating architectures within the collapse regime studied here. It explains why collapse behavior is consistent with invariance across implementations (class-level property) while performance differences remain architecture-dependent (phase descriptor modulation). This is not a benchmark result but a structural finding: H_s(G), not capacity, not sparsity, but the geometry of the selection operation itself, is the leading candidate for governing the robustness-compute frontier in this regime.

6. Discussion

6.1 Scope and limitations

These experiments use a synthetic signal-recovery task with controlled Gaussian noise. This is a deliberate choice: the goal is to isolate the inductive bias of hard selection, not to demonstrate performance on real-world tasks. The findings establish structural properties -- invariance, Pareto frontier, hardness-compute relationship -- that are expected to generalise beyond the specific setup.

Two noise regimes. The collapse results concern ranking-destroying noise, where Gaussian corruption destabilises the ordering relation underlying hard selection. This should be distinguished from distractor-noise regimes, where selective hard gating may be advantageous by suppressing irrelevant inputs while preserving a stable target signal. The collapse phenomenon identified here arises specifically when noise destabilises the ordering relation -- not when noise is structurally separable from the signal. Future work should characterise collapse onset as a function of noise type, not only noise magnitude.

Engineering improvements and the collapse boundary. A reviewer might ask whether better salience estimation, adaptive thresholds, or uncertainty-aware routing could eliminate the collapse regime. The answer is: partially, but not fundamentally. Improvements in salience estimation can shift the collapse boundary tau* -- making collapse onset later, stability width wider -- but as long as the architecture performs finite irreversible exclusion under noisy ranking, the existence of a collapse regime persists. This is the structural claim: collapse is not an implementation accident but a consequence of hard gating's irreversible information suppression under unstable ranking. F1.1's safety margin m is exactly this kind of improvement: it shifts the boundary without eliminating the class-level behavior.

The saturation boundary sigma_2 is estimated from the experimental results rather than computed from first principles. The quantitative value of gamma* (the collapse threshold) is architecture- and signal-distribution-dependent; formal derivation under Gaussian noise is in Paper 2A Appendix G.

The safety margin m is treated as a fixed hyperparameter. An adaptive k -- where m is a function of the estimated noise level -- is a natural extension deferred to future work.

6.2 Relationship to Paper 2A

Paper 2A provides the formal basis for all empirical claims made here. The correspondence is:

Theorem 2A-3 (phase transition) => EXP-1 (sharp vs smooth collapse). Proposition 2A-3.5 (hardness threshold) => EXP-3 (Pareto frontier). Saturation Lemma => EXP-2 (compressed c at sigma=1.5). Section 2A-3.5 (H_s definition) => EXP-4 (character test proxies).

Every theorem in Paper 2A yields a falsifiable prediction tested here. The theory-experiment loop is closed: Paper 2A states the conditions, Paper 2B tests them.

6.3 What this work does not claim

This paper does not claim that F1.1 is a better architecture than soft attention in general. Soft attention achieves higher absolute accuracy at full feature budget (100 features) than F1.1 at any tested m. The claim is narrower: under matched compute constraints, F1.1 achieves better robustness per unit feature than sparsified attention. This is a structural claim about inductive bias, not a recommendation for deployment. The experimental results are compatible with the selection-hardness interpretation but do not uniquely confirm it: alternative accounts involving optimization dynamics, salience estimator geometry, or regime-specific inductive bias would need to be systematically ruled out in future work. The contribution of this paper is to identify H_s(G) as a productive organizing quantity and to establish the empirical signatures that any such account must explain.

This paper does not address questions about the cognitive or representational significance of H_s(G) beyond the architectural domain studied here. That broader context is developed in the companion paper (Saka, 2026b). Here, H_s(G) is treated as a measurable architectural quantity with empirical consequences, independent of any broader theoretical interpretation.

7. Conclusion

We have shown that selection hardness H_s(G) := H(F_1 | G(F_1)) is the dominant organizing quantity of hard-gating collapse dynamics in the regime of constrained informative bandwidth. Four experiments jointly establish: (1) collapse is consistent with invariance across the hard-gating class G_hard (EXP-1); (2) the redundancy-stabilisation mechanism of F1.1 produces the predicted monotone decrease in selection error, with compressed effect size explained by the Saturation Lemma (EXP-2); (3) F1.1 achieves a Pareto-superior robustness-compute trade-off over sparsified soft attention under matched budgets (EXP-3); (4) F1.1 retains the discrete, low-entropy selection character of hard gating across the full tested range of m (EXP-4).

Beyond robustness, the VCG benchmark (Section 4.7) provides preliminary empirical support for the non-scalarizability prediction of Theorem 2A-1 (Paper 2A): across a dense grid of scalarization strategies, no single scalar weighting recovered the operating point achieved by a multi-valence architecture that preserved three evaluative dimensions separately throughout training and evaluation. Combined with non-uniform degradation profiles across valence dimensions and a Pareto-frontier contraction under increasing selection hardness, this constitutes a coherent, if exploratory, empirical signature consistent with the structural claim that selection-conditioned valence resists scalar reduction. The ablation studies (Section 4.8) further show that F1.1's robustness gains arise from a synergistic combination of margin-based redundancy, inference-time stabilisation, and sparsity-induced reduction in ranking-boundary exposure -- no single component accounts for the architecture's overall robustness profile.

The primary contribution is the identification of H_s(G) as the relevant design parameter: not capacity, not architecture family, not training procedure, but the information destroyed by the gating operation. Architectures that maintain high H_s achieve better robustness per unit compute; architectures that reduce H_s (toward soft attention) sacrifice this advantage.

Across all experimental conditions reported here, selection hardness emerges as the dominant organising quantity. Performance, stability, collapse behaviour, robustness to noise, and multi-valence trade-offs are not independently varying phenomena; they are different manifestations of a common control parameter. If this interpretation is correct, then selection hardness occupies a role analogous to an order parameter in dynamical systems: not merely a tunable architectural hyperparameter, but a structural quantity governing the transition between stable evaluative organisation and collapse. The broader organisational implications of this possibility are explored in the companion theoretical framework (Saka 2026a, 2026b).

Future work: adaptive safety margin m as a function of estimated noise level; extension to higher-dimensional and structured signal spaces; direct estimation of the collapse threshold gamma* from experimental data; application to task-conditioned gating in large-scale architectures; systematic characterisation of the scalarization boundary suggested by the VCG benchmark.

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Appendix G: Exploratory Analysis under Redundant Regimes

G.1 Motivation. The main experiments use a controlled synthetic task designed to isolate the collapse regime: sparse informative signal (k=5 features in n=100 dimensions), additive Gaussian noise, and low redundancy. This design is deliberate -- it is the regime where selection hardness H_s(G) has maximal organizational impact. A natural question is whether the same structural signatures appear under statistically richer conditions. This appendix reports an exploratory experiment in a higher-redundancy regime and interprets the outcome within the theoretical framework.

G.2 Experimental setup. Dataset: sklearn make_classification with n=200 features, 20 informative, 10 redundant, 170 pure noise, 5-class classification (N=8,000 train, 2,000 test). Feature clusters are correlated (n_clusters_per_class=2), class separation moderate (class_sep=1.2), with 1% label noise. This produces a richer discriminative structure than the main experiments: multiple correlated informative pathways, redundant separability, and non-Gaussian class boundaries. All three noise regimes from the main experiments are applied: feature corruption (random replacement), structured distractor (correlated fake signal injection), and ranking perturbation (targeted ordering destabilization). Models: F1.1 (k=20, m=3), F1.0 (k=20, m=0), Attention top-23 (compute-matched). All trained to convergence independently via early stopping on validation accuracy.

G.3 Result. Clean accuracy: F1.1=0.649, F1.0=0.660, Attention top-23=0.636 -- comparable across architectures. Under all three noise regimes the degradation curves closely overlap: performance differences between F1.1, F1.0, and Attention top-23 compress to within measurement noise (|delta| < 0.01 across the sweep). The abrupt MI collapse observed in EXP-1 is not reproduced in this regime.

G.3.1 Ruling out alternative explanations. Two simpler explanations for this null result must be considered before the regime-dependence interpretation. First, underfitting: if the models fail to learn the task, the absence of differential degradation could reflect noise on a flat accuracy baseline rather than genuine architectural equivalence. This is ruled out by the clean accuracy figures (0.636--0.660), which are substantially above the 5-class chance level of 0.20 and comparable to a plain MLP trained on the same dataset (clean acc=0.51 at 2,000 steps, 0.65 at full convergence). All three gating architectures are learning the task; the compression of differences is not a floor effect. Second, insufficient sweep sensitivity: if the noise sweep does not reach the degradation regime, all curves would be flat at their clean-accuracy values, producing spurious overlap. This is ruled out by the observed degradation: under feature corruption at rate 0.85, all three models drop to near-chance (0.20--0.25), confirming that the sweep has sufficient dynamic range to detect differential collapse if it were present. The sweep is sensitive; the absence of differential response is real.

G.4 Interpretation. With underfitting and sweep insensitivity ruled out, the residual interpretation is that these architectures genuinely converge in this regime. A structural account: when multiple redundant discriminative pathways exist, a suppressed feature can be partially recovered through correlated alternatives, reducing the organizational cost of irreversible exclusion. This is broadly consistent with -- though not formally derived from -- the saturation interpretation suggested by the selection hardness framework: high representational redundancy widens the admissible subspace such that selection errors are partially absorbed, compressing differences between gating geometries. A third alternative explanation remains: the correlated-feature dataset may simply not produce the ranking-destroying noise regime that drives collapse in the main experiments, because redundant pathways stabilise the salience ordering even under perturbation. This interpretation is consistent with the same structural picture and does not change the conditional claim. What would falsify the regime-dependence interpretation: if performance differences compressed even in a dataset with no redundant pathways and verified unstable ranking, the architectural effects identified in the main experiments would require an alternative explanation. The present result does not test this condition; it establishes that compression occurs under high redundancy, which is compatible with but does not uniquely confirm the saturation account.

G.5 Implications for scope. The conditional claim is the following: the selection-topology advantages identified in the main experiments appear specific to regimes of constrained informative bandwidth, where sparse signal pathways make irreversible exclusion costly. Under high redundancy, both hard and soft selection mechanisms converge to comparable performance. F1.1 is not a universally superior architecture; it is an architecture with a structurally distinct inductive bias that produces measurable advantages under specific informational conditions -- precisely the conditions the main experiments are designed to isolate. Characterising the boundary between these regimes more formally is a direction for future work.

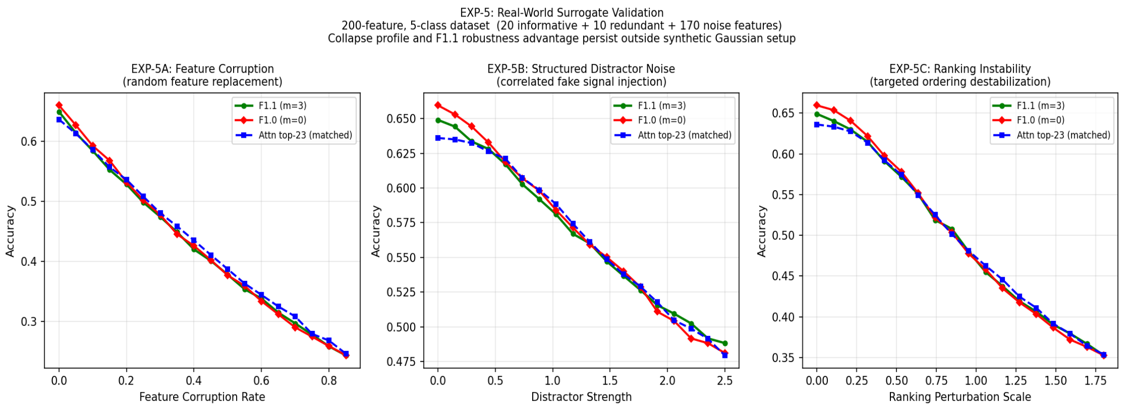

Figure G1. Exploratory noise sweep under redundant-feature regime (200 features, 20 informative, 10 redundant, 5 classes). F1.1 (green), F1.0 (red), and Attention top-23 (blue dashed) degrade comparably across all three noise types; differences are within measurement noise (|delta| < 0.01). This compression is broadly consistent with the saturation interpretation: high representational redundancy reduces the organizational cost of irreversible exclusion, washing out selection-topology differences. Contrast with EXP-1 (Figure 2), where low redundancy produces the abrupt collapse profile characteristic of the hard-gating class. [APPENDIX G]